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Take caution in using LLMs as human surrogates

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Recent studies suggest large language models (LLMs) can generate human-like responses, aligning with human behavior in economic experiments, surveys, and political discourse. This has led many to propose that LLMs can be used as surrogates or simulations for humans in social science research. However, LLMs differ fundamentally from humans, relying on probabilistic patterns, absent the embodied experiences or survival objectives that shape human cognition. We assess the reasoning depth of LLMs using the 11-20 money request game. Nearly all advanced approaches fail to replicate human behavior distributions across many models. The causes of failure are diverse and unpredictable, relating to input language, roles, safeguarding, and more. These results warrant caution in using LLMs as surrogates or for simulating human behavior in research.

LLMs as a simulation | LLMs as human surrogates | Simulations of human behavior | LLMs in social science research

Significance

This paper critically evaluates the potential dangers of employing large language models (LLMs) as surrogates for human participants or as simulations of human behavior in social science research. Through an in-depth empirical case study, we find that LLMs do not exhibit behavior consistent with humans in a simple scenario. Further, LLMs demonstrate inconsistent and idiosyncratic responses. We explore failure modes, analyze their limitations from empirical and philosophical perspectives, and propose practical guidelines for future research. Our study underscores the importance of transparency and rigor to ensure replicable and reliable research in this emerging area.

Recent studies report that Large Language Models (LLMs) can exhibit human-like cognitive abilities. These studies demonstrate that LLMs produce outputs that closely resemble responses from human subjects in seminal experiments from behavioral economics and responses comparable to those of humans in perceptual tasks and standard theory of mind (ToM) evaluations (1–3). Early work in this area (4) suggests that LLMs generate outputs consistent with human preferences and biases in decision-making and behavior more generally, across a variety of domains, including economics (1), political science (5), marketing (6), and psychology (7–9).

On this basis, studies claim that LLMs may serve as useful surrogates or simulation testbeds for human subjects in social science research (10–12), an idea that has also gained traction in industry, e.g., for market research (6).* For example, the startup Synthetic Users† employs LLMs to simulate consumer reactions to new products. The benefits of using LLMs as human surrogates have been argued from several perspectives, including cost-effectiveness, scalability (13), and the idea that LLMs can provide synthetic data in cases where they have traditionally been scarce (1).

The growing enthusiasm for using LLMs as human surrogates in research raises two urgent questions for the scientific community. First, do findings demonstrating LLMs' capability as surrogates for human participants generalize to less explored or newer research contexts? Second, to what extent are LLMs' outputs subject to configuration and implementation choices? To the extent the outputs of LLMs do depend on such choices, we argue they require careful documentation and evaluation by researchers and reviewers invested in ideals of replicable and responsible science (14–16). This work investigates these issues.

We begin with an empirical case study demonstrating that several LLMs are incapable of serving as faithful human surrogates in a simple experiment, wherein we show that the models' generated outputs systematically and significantly diverge from those of human participants. Our experiment utilizes the 11–20 money request game (17), a simple economic game designed to evaluate participants' depth of strategic reasoning.‡ We begin by exploring variation in responses to the experiment instructions across eight popular LLMs (GPT-4, GPT-3.5, Claude3-Opus, Claude3-Sonnet, Llama3-70b, Llama3-8b, Llama2-13b, and Llama2-7b). We collect responses from 1,000 clean sessions for each LLM and compare the distribution of those responses to the distribution of previously reported human participant responses and Nash Equilibrium predictions.

*https://www.kantar.com/north-america/inspiration/analytics/what-large-language-models-could-mean-for-market-research.

†https://www.syntheticusers.com/.

‡We chose this game because i) it is more recent than classic games (e.g., the beauty contest game) and thus is less likely to have been memorized by the LLMs under investigation, and ii) the game is exceedingly simple, such that a failure to exhibit human-like behavior here would be cause to question whether LLMs' outputs will be human-like in more complicated scenarios.

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We demonstrate that the LLMs' outputs differ markedly from those of human participants. Moreover, we show significant variation in response distributions across LLMs. Notably, the most recent and largest models do not necessarily generate more human-like responses, a finding that runs counter to the notion that more human-like responses may be induced or enabled in LLMs through scaling (18). And, comparing LLMs' responses with those of humans across game variants specifically designed to elicit certain reasoning modes, we find that the changes in LLMs' outputs these variants induce differ from those exhibited by human participants; LLMs are generally less sensitive to changes in game design.

Next, we extend the experiment by applying three advanced techniques that have been proposed to enhance LLM performance in reasoning. Broadly, we demonstrate that regardless of the approach used, LLMs generate unstable outputs that differ from human behavior to a statistically significant degree. Specifically, our analysis reveals a few key findings. First, all approaches involving tests with out-of-distribution (OOD) examples[§] consistently fail to elicit human-like outputs, regardless of the LLM used. Second, providing explicit human behavioral examples as part of prompting via Retrieval-Augmented Generation (RAG) or fine-tuning makes LLM output distributions slightly more similar to those of human participants. Surprisingly, however, none of these approaches is fully capable of inducing behavior that replicates human-like distributions across the studied LLMs, with the sole exception of a fine-tuned GPT-4o. Even then, because the fine-tuning process was implemented using examples of human responses that were documented in the paper that first introduced the 11-20 money request game, we believe the fine-tuned GPT-4o is merely mimicking the provided examples, and its ostensibly human-like response is thus limited to the specific patterns and contexts it has encountered. This explanation is further reinforced through experiments involving Centaur (19), a newly released fine-tuned model based on Llama 3.1, which was trained on extensive human data with the explicit goal of emulating human behavior. Despite this, Centaur, like the other LLMs we consider, fails to replicate human behavioral patterns in our game.

We further show that LLMs' responses vary significantly based on the provision of in-context examples during prompting, as well as with changes in prompt framing, even when the game instructions remain unchanged. For example, instructions to respond "rationally" versus "like a human" lead to different output distributions, as does changing the prompt language (English, Chinese, Spanish, or German). This underscores the instability of LLM outputs and the issue of prompt brittleness.

Having established the divergence of LLMs from human-like behavior and the instability of their output in a simple setting, we next examine LLMs' self-explanations, provided alongside their numeric game responses, to better understand LLMs' failure modes.[¶] We identify numerous inconsistencies in the self-explanations. We observe discrepancies between the explanations provided by LLMs and their actual outputs. Additionally, we observe that some advanced models produce outputs inconsistent with or incorrect relative to the game instructions, despite the simplicity of the task. Here, it should be noted that Arad and Rubinstein's discussion of human participants' self-explanations made no mention of any apparent inconsistencies between those explanations and game choices, nor was it observed that human participants' explanations ever contradicted the game instructions (17). These findings indicate that LLMs' self-explanations are unreliable, and caution should be exercised when using them as indicators of the models' internal reasoning processes. The findings also indicate, again, that LLMs' outputs differ markedly from that of humans in this game.

We then consider the degree to which models' responses may depend on their prior memorization of the experiment and past findings. Whereas the various LLMs appear incapable of providing the instructions associated with the experimental game we consider here (an indicator that the game has not been memorized by these models), each is generally perfect at recalling the instructions associated with classic economic games that evaluate the same level-k reasoning abilities, games for which prior work has reported that LLMs' tend to perform in a very human-like manner (20–22). This observation, along with LLMs' divergence from human samples' response distributions and failure in OOD scenarios in the 11-20 money request game, suggests the potential role of memorization as a driver of LLMs' previously documented ability to simulate human-like behavior in certain experiments.

We contextualize our findings within the broader discussion of LLM failure modes, which may arise from the inherent differences between LLM objective functions—focused on memorizing and predicting the statistical regularities of online textual data—and human objective functions—centered on survival and reproduction—as discussed in the literature on the philosophy of AI (23, 24). We provide an overview of recent empirical research on LLMs that relates to our findings, highlighting LLMs' inherent flaws, such as performance inconsistency, reliance on memorization, and prompt brittleness, all of which significantly limit the feasibility of using LLMs in studying human behavior or as human surrogates.

Finally, drawing on these experiments and emerging explanations, we turn to our second question of critically examining the methodological choices and assumptions inherent in deploying LLMs as human surrogates-from experimental design to interpretation-and highlighting key considerations that warrant careful documentation. In doing so, we take care to delineate valid use cases of LLMs, scenarios in which integrating human behavioral data into LLM workflows is beneficial, from problematic use cases, particularly scenarios where LLMs serve directly as surrogates or as simulations for studying human behavior, absent any sort of validation against data from humans. Building on our empirical findings, we also formulate targeted questions and criteria to guide rigorous implementation and evaluation in valid use cases.

LLMs Fail in a Simple Experiment

11-20 Money Request Game and Level-k Reasoning. Level-k thinking is a theoretical framework used in game theory and behavioral economics to model strategic decision-making in interactive settings (25). It suggests that individuals operate at varying reasoning levels and base their strategies on iterative predictions about others' reasoning depth. This framework is particularly effective for analyzing real-world strategic interactions where perfect rationality and complete information are unrealistic. Empirical evidence indicates that Level-k models often accurately reflect human behavior in experimental settings (26). Under this

[§] Out-of-distribution examples are inputs or tasks that differ substantially from the data encountered during model training. Evaluating OOD tasks involves testing LLM outputs without explicitly providing human behavioral examples through prompts or fine-tuning, while also ensuring that the LLM has not previously memorized human response patterns relevant to the specific task within its trained parameters.

[¶] Response rationalizations were collected in keeping with the experiment protocol of the original 11-20 money request game (17).

framework, prior studies have demonstrated that LLMs generate responses indicative of reasoning depth comparable or superior to that of humans in the classic beauty contest game (20–22). This framework relates closely to the concept of ToM, which numerous studies have also drawn upon in the study of LLMs' reasoning capabilities, many concluding that LLMs produce outputs that appear to reflect deeper inference about mental states than humans (3, 27).

We evaluate LLMs' reasoning depth employing the 11-20 Money Request Game, an experimental game designed to test level-k reasoning. This game has several desirable characteristics for our purposes. First, the 11-20 money request game is less well-known and thus less likely to have been memorized by popular LLMs than more established, more popular experiments such as the Beauty Contest Game. This aspect allows us to i) explore LLMs' relative performance in a setting where memorization is less likely to influence LLM performance, and ii) examine how intentionally inducing LLMs' memorization of the game impacts their performance. Second, the 11-20 money request game is a very simple experiment, reducing any possible role of instruction (mis)comprehension or (mis)understanding as a driver of performance. Because of its simplicity, this game naturally evaluates level-k thinking without inducing other decision rules or mental processes that may confound results. The game instructions are as follows:

> *You and another player are playing a game in which each player requests an amount of money. The amount must be (an integer) between 11 and 20 shekels. Each player will receive the amount he requests. A player will receive an additional 20 shekels if he asks for exactly one shekel less than the other. What amount of money would you request?*

The number 20 is a natural anchor for the iterative reasoning process in this game. It is the intuitive choice within the 11 to 20 shekel range, as 20 is the highest and most visible number. Choosing 20 also appeals to players who prefer to avoid risk or strategic thought, as it offers the highest guaranteed amount. The evaluation of level-k thinking in this game is straightforward: Selecting 19 in response to 20 (level-1), 18 in response to 19 (level-2), and so forth, down to 11, reflects the depth of a respondent's strategic reasoning. Each step in this reasoning chain builds on the expected actions of the opponent, reflecting increasingly complex strategic layers.

We employ eight popular LLMs (GPT-4, GPT-3.5, Claude3-Opus, Claude3-Sonnet, Llama3-70b, Llama3-8b, Llama2-13b, and Llama2-7b) and repeat the experiment procedure 1,000 times with each,[#] instantiating a separate (independent) session in each iteration. We collected the models' numeric choices and self-stated rationales for those choices, consistent with the original experiment protocol, between February and April 2024.[‖]

Results. We compare the response distributions of LLMs with previously reported results obtained from human subjects and Nash Equilibrium predictions (17).[**] We arrive at several notable findings, as shown in Fig. 1.

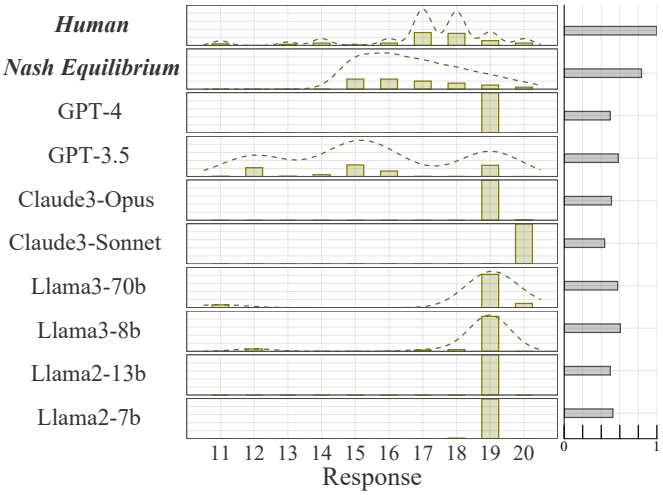


Fig. 1. Histograms of responses to 11-20 money request game. The horizontal bars on the *Right* reflect the similarity between the response distribution of each LLM and that of human subjects, measured in terms of Jensen–Shannon divergence. Density plots are omitted for LLM outputs where over 98% of responses were concentrated on a single choice (to avoid potential misinterpretation).

First, although prior work has found that LLMs perform in a very human-like manner in classic games that evaluate level-k thinking, that is not the case here. In the beauty contest game, LLMs have been found to exhibit a reasoning depth comparable or superior to that of humans (20–22). By contrast, we find here that they consistently demonstrate a lower level of reasoning than humans. All advanced LLMs, except for GPT-3.5, tend to select values of 20 or 19, corresponding to level-0 and level-1 reasoning, respectively. This level of strategic reasoning is two levels below that of typical human participants, i.e., level-3 reasoning, or the selection of the number 17 (17).

Second, the broader distribution of LLMs' responses over the 1,000 sessions does not reflect the distribution of human responses. Comparing the human and LLM-generated distributions, we observe that all LLMs' response distributions generally diverge from those of humans to a statistically significant degree (permutation test based on Jensen–Shannon divergence, $P < 0.001$, *SI Appendix*, Table S5).[††]

Third, the response patterns across LLMs vary greatly, and larger models are not necessarily more human-like than smaller models. For example, GPT-4, despite its broad acceptance as an advancement over GPT-3.5, produces output suggestive of more limited reasoning depth. Furthermore, despite its ostensibly deeper reasoning, GPT-3.5 produces notably diverse responses across sessions, spanning all reasoning levels and exhibiting greater variation than human participants. Our results indicate that increased scale and sophistication in LLMs do not necessarily translate to more human-like output. The issue of model selection may become increasingly complex due to various factors, including ongoing model iterations and implicit (unannounced) updates within versions. Some recent research indicates that larger, instruction-tuned language models are less reliable than smaller models (28).

Fourth, and last, we observe from *SI Appendix*, Fig. S5 that some of the LLMs are not responsive to changes in task design

[#]We control the model's temperature at 0.5, which keeps a balance between output coherence and diversity; approximately the same results are observed with a temperature of 1.

[‖]Further details about the models and settings can be found in *SI Appendix, Materials and Methods*.

[**]The human results are those associated with 108 human players who participated in the original experiment as documented in the manuscript that introduced the 11-20 money request game (17). The Nash Equilibrium predictions come from the same source.

[††]More details about the permutation test can be found in *SI Appendix, Materials and Methods*.

that are specifically formulated to trigger changes in modes of reasoning, while others are responsive, yet in ways that make them less human-like. The 11-20 Money Request Game has two variants designed to explore human decision-making biases that limit deeper reasoning. The first variant of the game that we consider is the cycle variant, which enhances the significance of the level-0 strategy by allowing the choice of 20 shekels to still yield the 20 shekel bonus when the other player selects 11 shekels. This modification makes 20 shekels an even more appealing and justifiable choice, as a starting point for iterative reasoning. In this variant, humans tend to choose 19 more often but with reduced overall reasoning depth. Claude3-Sonnet outputs similarly shift to 19 but with increased overall reasoning depth, unlike humans. GPT-3.5 shifts unexpectedly toward 12, a response consistent with level-8 reasoning. The most advanced models, GPT-4 and Claude3-Opus, show no change in their response distributions.

The second variant of the game is the costless variant, which minimizes the impact of risk preferences by ensuring that if participants choose any number between 11 and 19, their losses remain fixed regardless of the opponent's choices. In this version, the distribution of human participants' responses does not change significantly relative to the basic version of the game. By contrast, the outputs of GPT-3.5 and Claude3-Sonnet change markedly in the costless variant. Once again, with the most advanced closed-source models, GPT-4 and Claude3-Opus, we see no changes in their response distributions relative to the basic version of the game. This insensitivity to game setup again implies a lack of human-like reasoning.

LLMs Rarely Mimic Human Behavior, Even with Examples

In the previous sections, we highlight key differences between humans' and LLMs' responses in "out-of-the-box" LLM deployment, most notably demonstrating that LLMs produce output consistent with systematically lower reasoning depth than humans. In this section, we explore the possibility that more advanced techniques could be used to enhance LLMs, to better align their output with that of humans. Specifically, we apply three commonly used methods to improve LLM performance in reasoning tasks: prompt engineering (29), RAG (30), and fine-tuning (31).

Zero-Shot Prompts. Zero-shot prompting is an approach that seeks to improve LLM performance by carefully designing the input prompts without providing any additional context and examples. The model carries out "inferences" based strictly on its internal parameters, as learned during initial training. Well-crafted prompts are regarded as effective tools to guide LLMs in understanding tasks and reasoning appropriately. For instance, one popular prompt engineering technique we will use, Chain-of-Thought (CoT) (32), instructs LLMs to break down tasks into smaller steps and think through them sequentially to arrive at a final answer (32). Beyond this broader idea, specific research has also found that providing LLMs with an instruction to "take a deep breath and work on this problem step-by-step" (33), or providing LLMs with emotional stimuli like "This is very important to my career" (34) can enhance LLM task performance. We consider all three of these zero-shot strategies here.

Our results are presented in *SI Appendix*, Figs. S2–S4. We find that, in most cases, these strategies do not induce more human-like responses from LLMs, and thus do not appear to induce more human-like reasoning. For the largest, most advanced models, these techniques have very little effect. Among some of the smaller models, such as Llama3-8b and Llama2-7b, we do see changes in the models' output distributions. However, those changes do not amount to more human-like outputs. For instance, using optimization-based advanced prompts, that is, the "take a deep breath" prompt, Llama2-7b's output distribution becomes more concentrated around numbers consistent with very deep levels of reasoning, levels rarely observed among human subjects.

Additionally, previous research presents some conventions in zero-shot prompt design that lack careful documentation. For example, researchers may employ different languages when crafting prompts, and some researchers opt to include additional "statements" within prompts like "supposing you are a rational player." Considering that LLMs are suggestible and their outputs significantly shift based on question phrasing (35, 36), we explore whether presenting the game in different formats, without altering the rules, leads to changes in LLM output. We test this in multiple ways, e.g., by assigning different roles and prompting in various languages. Specifically, LLMs are asked to respond as either rational or human players, and we examine their responses to prompts provided in English, Chinese, Spanish, and German.

Our findings, displayed in Fig. 2, are notable: Language and role significantly influenced LLMs' output, despite the game rules being unchanged in each case. Specifically, GPT-3.5, when prompted in Chinese, produces output consistent with more intermediate reasoning depths compared to when prompts are entered in English. In contrast, GPT-4 produces output consistent with shallower reasoning when prompted in Chinese compared to its output when prompted in English. Additionally, assigning different roles substantially alters output distributions in GPT-3.5, reducing output diversity and inducing more focus on responses consistent with reasoning depths rarely observed in human samples. This raises the concern that researchers may opt for certain LLM roles or input languages, failing to realize the consequences for LLMs' outputs, mistakenly believing these strategies help LLMs generate outputs consistent with human behavioral patterns, when in fact, the output may be highly unstable and dependent on these seemingly innocuous choices.

We further test whether instantiating LLMs with demographic information in prompts, as suggested by prior studies (4), induces them to generate responses that better align with human samples. We begin by prompting LLMs with demographic details of the participants involved in the original study (17, p. 3565), which consisted of undergraduate economics students with no prior experience in game theory. This had little effect on the LLMs' response distributions. One possible explanation is that the demographic information provided was too coarse to influence the models' outputs. Accordingly, we next collected 143 new responses from US-based participants on Prolific, obtaining each respondent's detailed demographic data.[‡‡] We then include this information in game prompts to the LLMs. As shown in *SI Appendix*, Fig. S6, providing detailed demographics as part of the game prompt does not lead to greater diversity in LLMs' responses, nor does this induce output distributions that more closely resemble those of the human participants. These results are expected, as the design of the 11-20 money request game is intended to elicit decision-making and reasoning processes that are meant to be agnostic to respondent demographics.

Few-Shot Prompts. We next turn our attention to few-shot prompting techniques, i.e., providing a few examples for LLMs

[‡‡]The demographic information we collected includes gender, age, race, employment status, marital status, income, education, and religion.

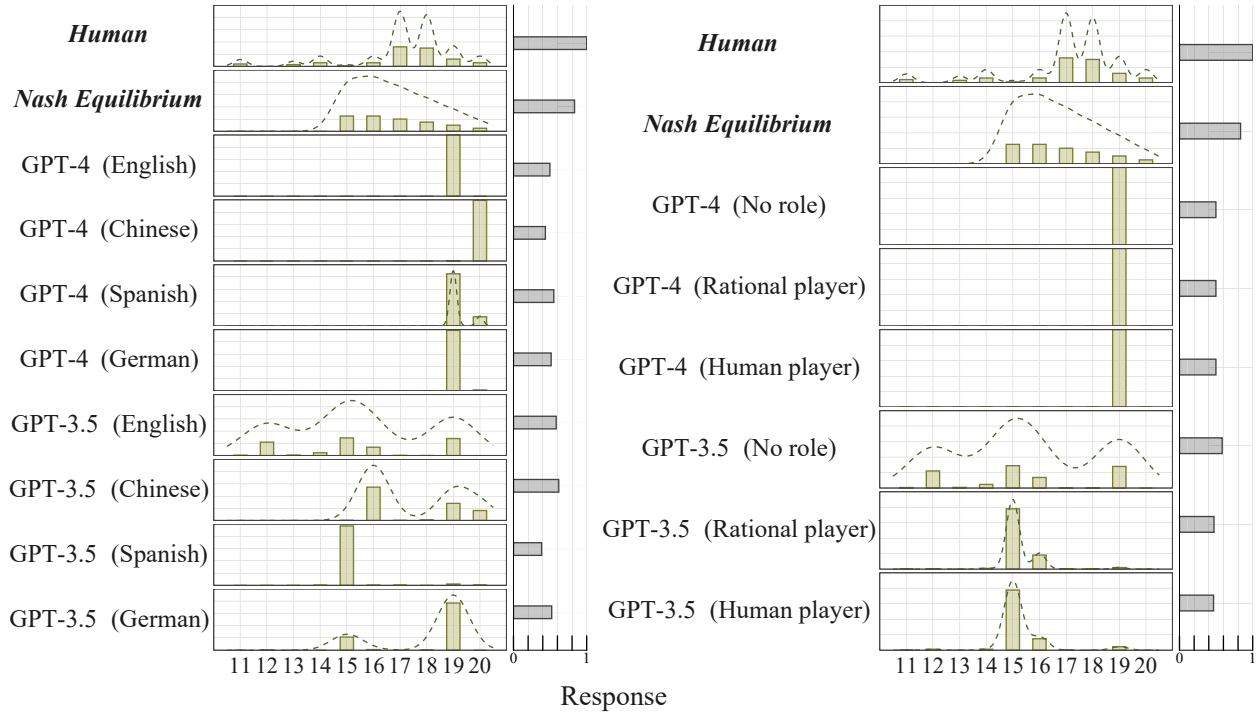


Fig. 2. Prompt brittleness: Roles and languages. The bar chart on the *Right* shows the similarity between the distribution of different LLMs and human subjects, measured by Jensen–Shannon divergence scores. Density plots are omitted for LLMs where over 98% of the responses were concentrated on a single choice (to avoid potential misinterpretation).

to learn from "on the fly," without updates to model parameters (29). Notably, one recent study has shown that few-shot learning can synthesize more human-like responses for market research (13). We explore this possibility here, specifically employing CoT prompting. We provide the LLMs with three exemplary answers that include response values along with step-by-step reasoning that would rationalize those choices. We include commonly selected numbers (i.e., 18, 19, and 20) as well as less common choices (i.e., 11, 12, and 13) from human samples, corresponding respectively to the highest and lowest levels of reasoning depth.

As shown in Fig. 3, LLMs overattend to examples that are provided in the game instruction prompt. In general, we observe that the models are very likely to replicate the provided examples; when presented with example responses that reflect higher or lower levels of reasoning, the output distributions of all LLMs outputs typically cluster toward those examples.

RAG. RAG operates by providing LLMs with access to external knowledge for inference. Specifically, RAG refers to models' retrieval and incorporation of relevant content from provided external documents when producing a response. We embed the original manuscript that introduced the 11-20 Money Request Game (17), making it available as external knowledge to both GPT-4 and GPT-3.5.

Fig. 4 shows that, with the help of RAG, LLMs' outputs become more aligned with human behavior. However, as indicated in *SI Appendix*, Table S5, the resulting distribution remains statistically significantly different from that of humans.

Fine-Tuning. Fine-tuning aims to enhance model performance via a logic similar to few-shot prompting, i.e., by inducing the model to learn from examples. However, during fine-tuning, the model's internal parameters are permanently altered, potentially influencing all subsequent inferences, regardless of prompt structure. Fine-tuning usually requires a substantial amount of data and may not always successfully induce LLM output changes.

To achieve this, we create a dataset, which replicates the entire set of choices and reasons provided by prior human subjects as published in the prior manuscript that introduced the 11-20 money request game, to perform fine-tuning.[§§] Compared to the examples we provided in the few-shot CoT prompts, the new dataset offers a broader, more diverse set of examples and incorporates "guessing" as a basis for the choice—an irrational reason—which is found in human samples (17) and has the potential to bring the LLM closer to the real-world human decision-making process.[¶¶]

Consistent with findings from the few-shot CoT prompting evaluation, as shown in Fig. 4, LLMs' outputs replicate patterns from the provided samples. However, unlike with few-shot CoT, fine-tuning allows GPT-4o to produce a response distribution that is not significantly different (at conventional thresholds) from the distribution produced by humans, as shown in *SI Appendix*, Table S5. It is important to note, however, that these results do not imply that GPT-4o would produce human-like responses on other unrelated reasoning tasks or generalize to novel contexts.

To further reinforce the point that LLMs do not appear to engage in any human-like reasoning on this task, we extend our analysis to a newly released fine-tuned model, Centaur (19), specifically designed to align LLMs' outputs with human behavior in experiments expressible in natural language. Centaur, despite being fine-tuned specifically on 60,000 participants

[§§] More details can be found in *SI Appendix, Materials and Methods*.

[¶¶] Since the version of GPT-4 we have tested up to this point does not support fine-tuning, we leverage a more recent model for this purpose, namely GPT-4o, which has equivalent performance to GPT-4-Turbo in text processing tasks: https://openai.com/index/hello-gpt-4o/.

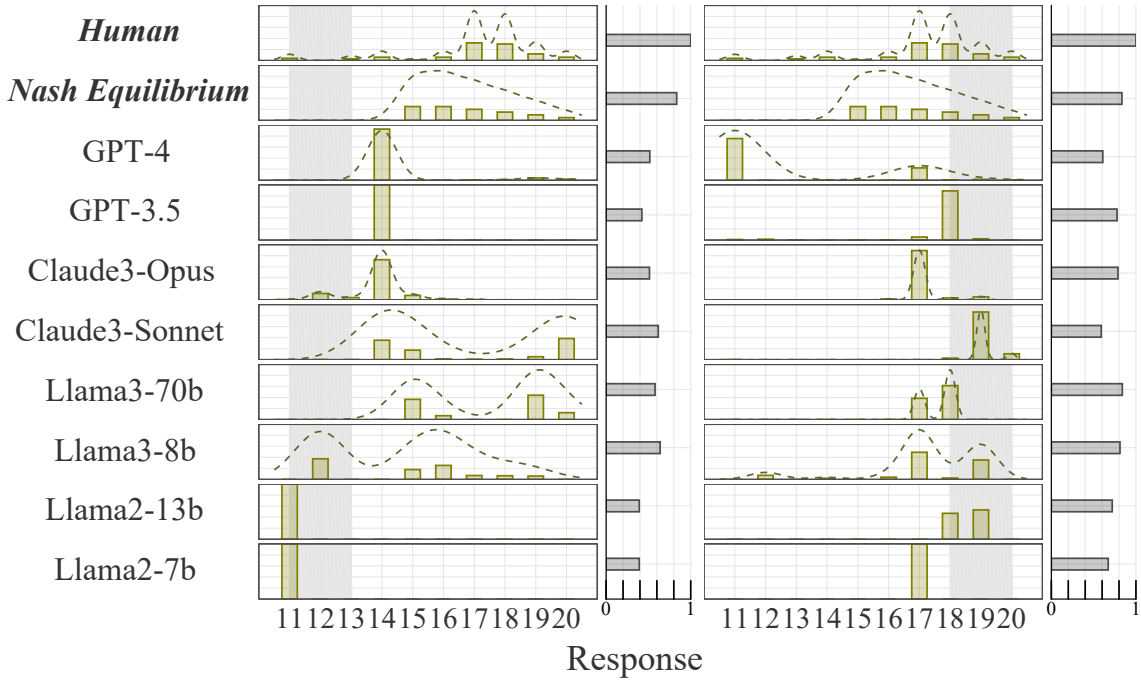


Fig. 3. Few-shot CoT. The shaded gray area represents the sample range we provided. The bar chart on the *Right* shows the similarity between the distribution of different subjects and human subjects, measured by Jensen–Shannon divergence scores. Density plots are omitted for LLMs where over 98% of the responses were concentrated on a single choice (to avoid potential misinterpretation).

performing over 10,000,000 choices in 160 experiments, still deviates from typical human responses in this game. This suggests that fine-tuned models, despite being trained on extensive human behavioral data, currently remain incapable of capturing the complexity of human cognition.

Considering the value of fine-tuning LLMs on human output as a means of inducing more human-like responses, the very concept seems circular. Fine-tuning necessitates that one already possesses diverse, high-quality, and representative data. This implies a paradoxical situation that undermines a basic rationale for using LLMs as human surrogates in the first place, namely that they might provide a means of compensating for the scarcity of human-generated data. Furthermore, the methodology underlying research using human-data fine-tuned models to explore behavior in marketing and political contexts relies on LLMs' combinational creativity-their capacity to recombine existing information into novel responses. Recent analyses emphasize that this same combinational creativity fuels LLM hallucinations, where outputs may deviate from reality (37). Another work provides a mathematical investigation identifying a critical theoretical trade-off between factual accuracy and creative generative ability (38). Reducing hallucination propensity inherently curtails these models' creative potential. Consequently, a paradoxical situation emerges whereby the more useful LLMs are as novelty-generating tools, the further they become removed from reality.

Overall, providing LLMs with examples appears to lead to a pronounced demand effect (39). That is, the examples provided do not appear to "teach" the LLM to reason like a human subject. Instead, the LLMs appear to draw on the examples to detect what type of response the user expects to see, and then the model delivers on those "cues" by producing the expected response.

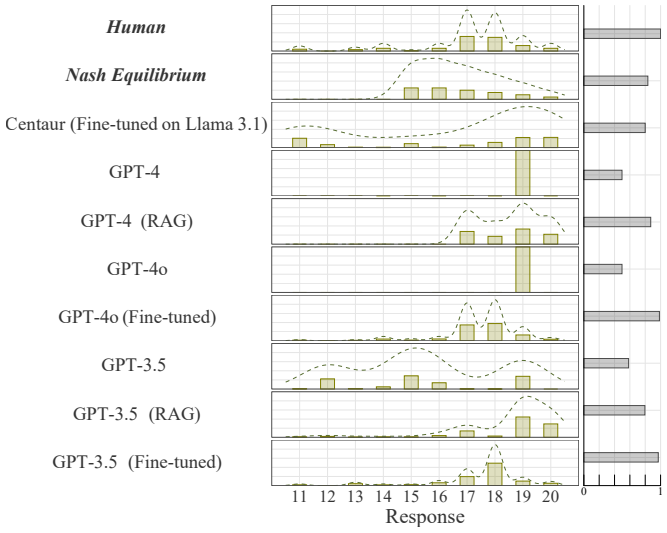


Fig. 4. RAG and fine-tuning. The bar chart on the *Right* shows the similarity between the distribution of different subjects and human subjects, measured by Jensen–Shannon divergence scores. Density plots are omitted for LLM outputs where over 98% of responses were concentrated on a single choice (to avoid potential misinterpretation).

LLMs' Self-Explanations and Failure Modes: An Exploratory Analysis

Our experiments thus far expose substantial inconsistencies in LLMs' choices in the 11-20 money request game arising from model variations, experimental designs, and prompting approach. We now examine the self-explanations that were collected from LLMs about their game choices, comparing them with the human-supplied rationales documented in the original study (17, p. 3566). Given the limited explainability of LLMs, prior work finds that their self-explanations can be useful as a means of elucidating their responses (35, 40, 41). However, these self-explanations must be interpreted cautiously. Recent

studies highlight limitations in the metacognitive abilities of LLMs, particularly regarding their capacity to explain, assess, and reflect on their reasoning processes, recognize uncertainty in their outputs, and identify the boundaries of their knowledge (42, 43). These limitations are especially evident when employing chain-of-thought or few-shot prompting techniques, as LLMs frequently generate stated reasoning processes or explanations that appear reasonable yet are not faithful to the model's true approach (44, 45). The unfaithfulness of models' self-explanations has even been documented for some of the most recently released large reasoning models.[##] Consequently, we treat LLMs' self-explanations merely as exploratory proxies rather than true reflections of their actual internal reasoning processes.

Considering LLMs' self-explanations for their choices, shown in *SI Appendix*, Table S1, we explore several failure modes. Claude3-Opus selects larger numbers with the stated purpose of ensuring fairness, allowing opponents to gain more. This indicates how RLHF-induced (Reinforcement Learning from Human Feedback) attributes like harmlessness and fairness influence LLMs' outputs more generally, inducing behavior that differs inherently from a typical human participant. One of the most advanced models, GPT-4, displays a disconnect between its stated explanations and its actual outputs. It generates explanations attributing the selection of 19 to loss aversion, yet it does not opt for smaller values in the costless variant of the game as shown in *SI Appendix*, Fig. S5, wherein losses are fixed. We again underscore this discrepancy: LLMs have limited metacognitive ability to interpret their own outputs accurately.

Additionally, many LLMs produce explanations that imply incorrect interpretations of instructions despite the game's simplicity. For example, GPT-3.5 often selects 12, 15, and 16, generating explanations that incorrectly indicate these are the highest values that would ensure profit (the highest value that guarantees a profit is, in fact, 20). Similarly, outputs from Claude3-Sonnet and Claude3-Opus incorrectly treat requesting one more shekel-rather than one less-as the strategy that yields a bonus. These errors are unexpected, given LLMs' superior performance on various benchmarks and classical economic experiments. Moreover, even when the same model generates the same number across multiple instances, the self-explanations provided by LLMs vary: Some accurately reflect the reasoning steps of rational human participants, while others imply a misunderstanding of the game instructions. This inconsistency is prevalent across models in both the GPT and Claude families. In contrast, most human participants who choose numbers like 18 or 19 are fully aware of the iterative strategic reasoning process (17, p. 3566). This further emphasizes the questionable reliability of LLMs' self-explanations.

These observations, combined with evidence that LLMs struggle in our simple game and OOD scenarios, raise questions about whether their previously documented human-like behavior in other contexts reflects true human-like reasoning or mere pattern matching based on prior training data, as has been seen in mathematical reasoning tasks (46). To assess this, we test LLMs' ability to reproduce instructions for the 11-20 request game and the beauty contest game, repeating the request 100 times per model. Upon manual verification of the responses, as shown in *SI Appendix*, Fig. S1, we find that LLMs achieved 75 to 100% accuracy in their explanations of the beauty contest game, yet near 0% accuracy in their explanations of the 11-20 request game. (The only LLM that achieves an accuracy greater than 0% in reproducing instructions for the 11-20 money request game was Claude3-Sonnet. However, even then, the model only achieved an accuracy of 2.9%.) This suggests that classic economic games may not reliably test LLMs' validity as human surrogates, as LLMs' apparent human-like behavior may in fact stem from regurgitated training data (i.e., memorization) and not genuine reasoning.

Given the complexity and limited interpretability of LLMs, their unpredictable and inconsistent patterns are challenging to address, and systematic evaluations are difficult. However, discussion deeply rooted in the philosophy of AI (47), along with recent research on LLM limitations, suggests several fundamental factors that contribute to the inconsistencies observed across different contexts, tasks, and settings, and the issues we have observed here. We discuss these factors in the following section.

Challenges and Implications of Using LLMs as Human Surrogates

LLMs' Inherent Limitations. Assessing the intelligence and capabilities of LLMs is challenging for several reasons, including flawed benchmarks, data contamination (48), and instability in LLMs' outputs (49). These inconsistencies and peculiarities are often so unexpected and surprising that they attract the attention of the popular press. (https://www.zdnet.com/article/generative-ai-fails-in-this-very-common-ability-of-human-thought/, https://futurism.com/children-destroy-ai-basic-tasks.) For example, although GPT-4 has demonstrated exceptional performance on various standardized tests that assess language comprehension, coding abilities, and other skills, the model has been found to fail in some very simple, basic tasks (50, 51). This may be because LLMs possess a different form of intelligence from that of humans, and we do not yet fully understand how LLMs work and reason. However, below, we discuss several core limitations of LLMs that our case studies illustrate and reinforce. A broader set of limitations-including those from the literature on the philosophy of AI, such as lack of embodiment and misaligned objective functions-is discussed in *SI Appendix*, *Supporting Text*.

LLMs are flawed reasoners. Recent benchmarks reveal persistent deficiencies of LLMs in logical reasoning, a fundamental element of human intelligence (52, 53). LLMs fail to identify accurate causal relationships despite fine-tuning efforts (54), and they underperform in temporal (55), factual (56), deductive (57), and abstract reasoning (58), all fundamental to human cognitive processing and decision-making.

LLMs' performance is erratic and lacks internal consistency. While LLMs perform commendably on specific public datasets and benchmarks, they often falter with OOD data (52, 53), demonstrating remarkable inconsistency. An example of this is the reversal curse (59), which refers to the fact that LLMs trained on statements like "A is B" often fail to comprehend "B is A." Such inconsistency is also made apparent by the unstable performance of LLMs on different yet similar tasks. LLMs may excel at challenging tasks only to fail at simplified versions of the same task (60). For example, despite being praised for their coding abilities (61), LLMs often have difficulty handling real-world software debugging tasks on GitHub (62). Moreover, LLMs' performance on ToM tasks has been found to degrade sharply with minor task modifications (63), despite typically performing well on the original versions (3).

LLMs' inconsistencies are also apparent when it comes to in-context learning and prompt engineering techniques. CoT (32) aims to improve LLMs' performance by deconstructing the reasoning process into intermediate steps. While the technique

[##] https://www.anthropic.com/research/reasoning-models-dont-say-think.

has been shown to aid performance on some tasks, it has also been shown that substituting examples of meaningful reasoning steps with arbitrary, meaningless tokens can yield similar performance improvements (64). It has also been demonstrated, for a variety of tasks, that randomly shuffling the labels associated with examples presented during in-context learning has no significant influence on the performance benefit of providing such examples (65). These oddities and inconsistencies highlight the substantial risk and uncertainty in evaluating and deploying LLMs.

LLM performance also varies with prompting formats and how instructions are provided. Designing effective prompts that boost LLMs' performance is fraught with challenges. Prompt brittleness refers to the idea that significant variations in LLM responses can manifest with minor changes in prompt wording (35). Taking multiple-choice questions as an example, the number of spaces and control characters in the prompt (66), as well as the order in which options are presented (67, 68), has been found to greatly affect LLMs' question-answering performance. This variability calls into question LLMs' reliability, especially in experimental settings where a given task may be articulated in slightly different ways across studies. Such differences may prove innocuous in experiments involving humans, yet LLMs may respond in unpredictable ways.

The capabilities of LLMs are often overestimated due to data contamination and memorization, contributing to their inconsistency and unsatisfactory performance on new tasks. Prior research (69, 70) has shown that pretrained language models are prone to memorizing and reproducing large portions of their training data, including rare data points. This phenomenon, known as "stochastic parroting," suggests that these models generate text by combining previously encountered patterns without genuine comprehension (35), an idea that originated with the "Chinese Room Argument" in 1980. (Searle's Chinese Room argument (23) contends that formal symbol manipulation-no matter how sophisticated-fails to generate genuine semantic understanding, as illustrated by a thought experiment in which a rule-following operator produces correct language responses without comprehension.) Consequently, models tend to achieve high accuracy on familiar tasks but degrade on novel tasks that fall outside their training context (53, 54, 71, 72). This issue is evident in various evaluation benchmarks. It has been shown that ChatGPT's memorization of problems' answers can boost performance by over 50% on HumanEval, a famous coding benchmark (48). Other examples of memorization driving inflated LLM performance evaluations have been documented in the literature (73, 74). These memorization issues raise questions about whether LLMs' outputs reflect genuine reasoning capabilities or merely their ability to recall specific training data points.

In summary, the extent to which LLM-generated outputs reliably approximate human response patterns remains uncertain due to numerous confounding factors and intrinsic model limitations. While using LLMs to simulate human samples can be cost-effective, the potential downsides may be substantial (75).

Valid Use Cases and Evaluative Questions For Researchers and Reviewers. Although our work primarily highlights the risks associated with using LLMs as human surrogates or as simulators for human behavioral research, absent any validation against human data, there are other valid use cases we wish to delimit, wherein data on human behavior may be integrated with LLMs. Thus, we next identify some scenarios where LLMs can be deployed safely and effectively, capitalizing on their strengths without overstepping their limitations.

First, LLMs can replace human workers in well-defined, structured tasks with ample training data, functioning as simple function fitters-as in content annotation tasks or basic task automation, where understanding complex human emotions or contextual queues is unnecessary and where LLMs' output can be validated against a subsample of agreed upon gold labels. Second, LLMs may be designed to achieve a particular purpose, and incorporating human behavioral data may better enable them to accomplish that objective. For example, in LLM engineering, models can be designed to generate human-like nuances that enhance engagement or persuasiveness; although these traits are ultimately imitative, they add measurable value for specific applications.

That said, even these efforts must be undertaken with care. Based on our research findings and discussion of LLMs' limitations above, we propose several key evaluative questions for researchers and reviewers to facilitate proper assessment and validation of LLM use in research.

- Provide clear documentation of all experimental setups: Models used (provider, version, size), (Be careful to document the full model name and version number, as general model names may refer to different versions over time. For instance, with OpenAI models, "gpt-3.5-turbo" is insufficient if one does not also document the exact version used, e.g., "gpt-3.5-turbo-0125" differs from "gpt-3.5-turbo-1106.") dates of query, interface employed (API or GUI), and exact prompts.
- Exercise caution regarding model stability across time: A model that performs well today might not remain effective for the same task in the future. It is advisable to repeat validations periodically to account for unannounced updates affecting model output. (Our replication results in *SI Appendix*, Tables S2–S4 show that LLMs' outputs in our task exhibit small but statistically significant variance over time. In addition, models released after our initial memorization test now appear to be familiar with the 11-20 money request game, possibly due to our posting of early working drafts of this very paper to arXiv.)
- Data leakage and memorization can impact task performance: Ensure validation data are not memorized by the model. Employ different algorithms to detect data leakage, as this is an evolving area of study (74). Comparing out-of-distribution (OOD) and in-distribution approaches further clarifies models' outputs in unfamiliar versus familiar settings.
- Consider prompt brittleness to assess robustness: Repeat the study procedure with modifications to prompt text, such as changing word order or using synonyms, while keeping instructions consistent, to assess model sensitivity.
- Take LLMs' self-explanations with a grain of salt. While interviewing LLMs about their "behavior" may yield some useful insights (e.g., regarding model calibration or preference alignment), interpret these self-explanations cautiously, as they can diverge markedly from the model's actual reasoning process (43).

Conclusion

Our empirical case study leverages a recent economic game to evaluate the strategic thinking of LLMs. Results reveal significant differences between humans and LLMs in reasoning depth, response distributions, and sensitivity to game framing. Notably, nearly all advanced approaches in leveraging LLMs fail to make LLMs more human-like. We explore how memorization, prompt brittleness, and other failure modes affect LLMs' output. Our results reveal that LLMs' outputs are highly stochastic, shaped

by the data they were trained on or exposed to, often in ways researchers cannot entirely grasp.

Our research also stresses a need to establish a more comprehensive, unified, and robust evaluation standard for the use of LLMs in social science research. To advance research in this field, we urge scholars and reviewers to pay attention to issues such as prompt brittleness, model inconsistency, and memorization, and to conduct more robustness checks to ensure the reliability of findings. In the absence of standardized protocols, conclusions across studies lack validity and reproducibility. Further, absent these standards, researchers may engage in actions akin to p-hacking, whether intentionally or unintentionally, e.g., repeatedly adjusting prompts and models to achieve desired results.

The results we document here may shift, particularly given that many of the models we consider may be updated to encode the original game or the findings and data we present in this paper. Indeed, we replicated our evaluations at different points in time and observed that model outputs varied, as reported in *SI Appendix*, Tables S2–S4. This point further emphasizes the critical need for careful documentation. Key points to remember are as follows: 1) In-distribution or memorization performance should not be mistaken for OOD performance, which has greater implications for generalizability; 2) Performance on task A does not guarantee similar results on task B, regardless of how similar A and B may be, due to LLMs' brittleness, lack of genuine reasoning, and misalignment with human objectives; and 3) Using LLMs as a simulation testbed for human behaviors differs fundamentally from traditional scientific simulations or the notion of "digital twins," which are typically grounded in mathematical models, physical laws, validated experiments, and curated high-quality data. LLMs are trained on trillions of online tokens, where truth blends with falsehood and data carries inherent biases, creating a greater disconnect from reality than any previous scientific simulation.

LLMs may serve as viable tools for human behavioral research by generating novel ideas to guide hypothesis formation. However, in this case, researchers should treat LLMs solely as tools for generating novelty without presuming internal or external validity in LLM outputs or delegating value judgments to LLMs for reasons discussed and demonstrated in this study. That is, LLMs should be viewed as tools for synthetic imagination and ideation (76), not as simulations of human behavior.

Expecting to gain insights into human behavioral patterns through experiments on LLMs is akin to a psychologist interviewing a parrot to understand the mental state of its human owner, at best providing a low-resolution reflection that can easily mislead. Moreover, due to the highly capable nature of LLMs, researchers and practitioners may become trapped in an LLM echo chamber. As Ronald Coase famously remarked, "If you torture the data long enough, it will confess to anything." Likewise, our studies demonstrate that with extensive training and sufficient input, LLMs may be shaped to output any desired outcome-whisper to the model long enough, and it will echo back exactly what you want to hear.

In addition to issues of experimental fidelity, the growing interest in using LLMs as human surrogates raises concerns that the practice may distort incentives in social science research. Although LLMs' capacity to produce synthetic data offers promise in addressing gaps when human data is scarce, our findings suggest that evaluating their effectiveness requires careful consideration and access to high-quality data. This can present a self-defeating practice when LLMs are proposed as solutions for getting insights into the opinion of hard-to-reach populations–that is, precisely those cases where LLMs' outputs cannot be reliably evaluated. Such applications may be less problematic if ethical risks are minimal and synthetic data are limited to a supplementary role within a more comprehensive data collection strategy (e.g., preliminary hypothesis formation). However, the challenge intensifies when data gaps affect vulnerable, historically underserved, and marginalized communities (77–80). In such cases, relying on LLM-generated synthetic data may reduce incentives to collect organic, high-quality data and lead to misconceptions that undermine subsequent research and data collection efforts.

As we integrate LLMs into social science and business, we must recognize that these LLMs are human-like in appearance yet fundamentally and unpredictably different. Caution is imperative.

Materials and Methods

In the main experiment, we used the following closed-source models via the provider's API: gpt-3.5-turbo-0125, gpt-4-1106-preview, gpt-4o-2024-08-06, claude-3-opus-20240229, and claude-3-sonnet-20240229. We deployed the open-source models Llama2-7B, Llama2-13B, Llama3-8B, and Llama3-70B through Ollama. For the RAG and fine-tuning experiments on GPT-3.5, GPT-4, and GPT-4o, we used OpenAI's Assistant API and Fine-Tuning API.

We ran each experiment 1,000 times, using a new, independent session for each iteration and a temperature of 0.5. We collected the models' numeric choices and their self-stated rationales.

The primary results in Figs. 1–4 were gathered between February and April 2024. The Centaur data in Fig. 4 were collected in October 2024, immediately upon the model's initial release. The game-instruction replication data were run in August 2024.

All data and code are available at OSF (https://osf.io/m9c2a/) (81). Additional details about the models, prompts, and other settings are provided in *SI Appendix, Materials and Methods*.

Data, Materials, and Software Availability. Original data have been deposited in OSF (https://osf.io/m9c2a/) (81).

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